EXPERIMENT - 7

7. Perform basic video processing operations on the captured video. Read captured video in python and display the video, in slow motion and in fast motion.

Aim

To read a captured video in Python using OpenCV and display the video in normal speed, slow motion, and fast motion.

Procedure

1. Import the OpenCV library.
2. Open the captured video using cv2.VideoCapture().
3. Read and display each video frame.
4. For slow motion, increase the frame display delay.
5. For fast motion, decrease the frame display delay.
6. Press the Esc key to stop the video.

Program

```
import cv2

cap = cv2.VideoCapture("video.mp4")

while cap.isOpened():
    ret, frame = cap.read()
    if not ret:
        break
    cv2.imshow("Normal Video", frame)
    if cv2.waitKey(30) & 0xFF == 27:
        break

cap.release()

cap = cv2.VideoCapture("video.mp4")

while cap.isOpened():
    ret, frame = cap.read()
    if not ret:
        break
    cv2.imshow("Slow Motion", frame)
    if cv2.waitKey(100) & 0xFF == 27:
```

```
        break

cap.release()

cap = cv2.VideoCapture("video.mp4")

while cap.isOpened():
    ret, frame = cap.read()
    if not ret:
        break
    cv2.imshow("Fast Motion", frame)
    if cv2.waitKey(1) & 0xFF == 27:
        break

cap.release()
cv2.destroyAllWindows()
```

Sample Input

Video.mp4

Sample Output

- Normal Video: Video plays at approximately normal speed.
- Slow Motion: Frames are displayed with a longer delay.
- Fast Motion: Frames are displayed with a shorter delay.

Result

Thus, the captured video was successfully read and displayed using OpenCV, demonstrating normal, slow-motion, and fast-motion playback.